Feedback

Workshop 6

Workshop 6 feedback

- Overloading resources makes everything run slowly.

- In the workshop we saw how to use the queuing system to run OpenMP, MPI and hybrid (MPI/OpenMP) programs

Workshop 6

OpenMP

```
#!/bin/bash
# Number of nodes requied
#SBATCH --nodes=1
# Number of tasks per node
#SBATCH --ntasks-per-node=1
# Number of CPUs per task
#SBATCH --cpus-per-task=8
# Maximum time the job will run for (HH:MM:SS)
#SBATCH --time=1:00
# Name of the job
#SBATCH --job-name=Hello_OpenMP
# Queue for the job to run in: pq is the parallel queue
#SBATCH -p pq
# Specify the account code and reservation
#SBATCH -A undergraduate_teaching-ecmm461
#SBATCH --reservation=undergraduate_teaching_2021

# Unload all exisiting modules
module purge
# Load the Intel Cluster Toolkit module used to compile the program
module load intel/2017b

# Run the program
./hello

# End of file
```

1 compute node

1 process

8 OpenMP threads

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Question 1: How did you modify the job script to change the number of OpenMP threads from 8 to 16?

```
#!/bin/bash
# Number of nodes requied
#SBATCH --nodes=1
# Number of tasks per node
#SBATCH --ntasks-per-node=1
# Number of CPUs per task
#SBATCH --cpus-per-task=16
# Maximum time the job will run for (HH:MM:SS)
#SBATCH --time=1:00
# Name of the job
#SBATCH --job-name=Hello_OpenMP
# Queue for the job to run in: pq is the parallel queue
#SBATCH -p pq
# Specify the account code and reservation
#SBATCH -A undergraduate_teaching-ecm3446
#SBATCH --reservation=undergraduate_teaching_2021

# Unload all exisiting modules
module purge
# Load the Intel Cluster Toolkit module used to compile the program
module load intel/2017b

# Run the program
./hello

# End of file
```

16 OpenMP threads

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Question 2: How many MPI processes are started by the `mpirun` command and where do they run?

MPI

```
#!/bin/bash
# Number of nodes requied
#SBATCH --nodes=2
# Number of tasks per node
#SBATCH --ntasks-per-node=8
# Number of CPUs per task
#SBATCH --cpus-per-task=1
# Maximum time the job will run for (HH:MM:SS)
#SBATCH --time=1:00
# Name of the job
#SBATCH --job-name=Hello_OpenMP
# Queue for the job to run in: pq is the parallel queue
#SBATCH -p pq
# Specify the account code and reservation
#SBATCH -A undergraduate_teaching-ecmm461
#SBATCH --reservation=undergraduate_teaching_2021

# Unload all exisiting modules
module purge
# Load the Intel Cluster Toolkit module used to compile the program
module load intel/2017b

# Run the program
mpirun ./hello_mpi

# End of file
```

2 nodes

8 processes per node

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```
Hello World from process 8 of 16 running on comp002
Hello World from process 0 of 16 running on comp001
Hello World from process 12 of 16 running on comp002
Hello World from process 4 of 16 running on comp001
Hello World from process 6 of 16 running on comp001
Hello World from process 1 of 16 running on comp001
Hello World from process 10 of 16 running on comp002
Hello World from process 2 of 16 running on comp001
Hello World from process 14 of 16 running on comp002
Hello World from process 3 of 16 running on comp001
Hello World from process 9 of 16 running on comp002
Hello World from process 5 of 16 running on comp001
Hello World from process 11 of 16 running on comp002
Hello World from process 13 of 16 running on comp002
Hello World from process 15 of 16 running on comp002
Hello World from process 7 of 16 running on comp001
```

Processes 0-7 run on **`comp001`**
Processes 8-15 run on **`comp002`**

These are the compute node hostnames

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Question 3: How would you modify the job script to run 32 MPI processes on 2 nodes with 16 processes on each node?

```
#!/bin/bash
# Number of nodes requied
#SBATCH --nodes=2
# Number of tasks per node
#SBATCH --ntasks-per-node=16
# Number of CPUs per task
#SBATCH --cpus-per-task=1
# Maximum time the job will run for (HH:MM:SS)
#SBATCH --time=1:00
# Name of the job
#SBATCH --job-name=Hello_OpenMP
# Queue for the job to run in: pq is the parallel queue
#SBATCH -p pq
# Specify the account code and reservation
#SBATCH -A undergraduate_teaching-ecm3446
#SBATCH --reservation=undergraduate_teaching_2021

# Unload all exisiting modules
module purge
# Load the Intel Cluster Toolkit module used to compile the program
module load intel/2017b

# Run the program
mpirun ./hello_mpi

# End of file
```

Increase number of tasks per node to 16

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Question 4: How many MPI processes and OpenMP threads are there and where do they run?

Hybrid

```
#!/bin/bash
# Number of nodes requied
#SBATCH --nodes=2
# Number of tasks per node
#SBATCH --ntasks-per-node=2
# Number of CPUs per task
#SBATCH --cpus-per-task=8
# Maximum time the job will run for (HH:MM:SS)
#SBATCH --time=1:00
# Name of the job
#SBATCH --job-name=Hello_OpenMP
# Queue for the job to run in: pq is the parallel queue
#SBATCH -p pq
# Specify the account code and reservation
#SBATCH -A undergraduate_teaching-ecm3446
#SBATCH --reservation=undergraduate_teaching_2021

# Unload all exisiting modules
module purge
# Load the Intel Cluster Toolkit module used to compile the program
module load intel/2017b

# Run the program
mpirun ./hello_hybrid

# End of file
```

2 nodes

2 MPI processes per node

8 OpenMP threads per node

```
[0] MPI startup(): Rank    Pid     Node name  Pin cpu
[0] MPI startup(): 0       1625    comp001    {0,1,2,3,4,5,6,7}
[0] MPI startup(): 1       1626    comp001    {8,9,10,11,12,13,14,15}
[0] MPI startup(): 2       9314    comp002    {0,1,2,3,4,5,6,7}
[0] MPI startup(): 3       9315    comp002    {8,9,10,11,12,13,14,15}
[0] MPI startup(): I_MPI_DEBUG=5
```

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```
Hello World: process 0 of 4, thread 0 of 8, running on comp001
Hello World: process 0 of 4, thread 7 of 8, running on comp001
Hello World: process 0 of 4, thread 6 of 8, running on comp001
Hello World: process 0 of 4, thread 1 of 8, running on comp001
Hello World: process 0 of 4, thread 3 of 8, running on comp001
Hello World: process 0 of 4, thread 5 of 8, running on comp001
Hello World: process 0 of 4, thread 4 of 8, running on comp001
Hello World: process 2 of 4, thread 0 of 8, running on comp002
Hello World: process 2 of 4, thread 4 of 8, running on comp002
Hello World: process 2 of 4, thread 2 of 8, running on comp002
Hello World: process 1 of 4, thread 0 of 8, running on comp001
Hello World: process 1 of 4, thread 1 of 8, running on comp001
Hello World: process 1 of 4, thread 2 of 8, running on comp001
Hello World: process 1 of 4, thread 6 of 8, running on comp001
Hello World: process 1 of 4, thread 3 of 8, running on comp001
Hello World: process 1 of 4, thread 5 of 8, running on comp001
Hello World: process 1 of 4, thread 4 of 8, running on comp001
Hello World: process 1 of 4, thread 7 of 8, running on comp001
Hello World: process 3 of 4, thread 0 of 8, running on comp002
Hello World: process 3 of 4, thread 1 of 8, running on comp002
Hello World: process 3 of 4, thread 3 of 8, running on comp002
Hello World: process 2 of 4, thread 6 of 8, running on comp002
Hello World: process 3 of 4, thread 7 of 8, running on comp002
Hello World: process 3 of 4, thread 5 of 8, running on comp002
Hello World: process 2 of 4, thread 7 of 8, running on comp002
Hello World: process 2 of 4, thread 5 of 8, running on comp002
Hello World: process 3 of 4, thread 2 of 8, running on comp002
Hello World: process 2 of 4, thread 3 of 8, running on comp002
Hello World: process 3 of 4, thread 4 of 8, running on comp002
Hello World: process 2 of 4, thread 1 of 8, running on comp002
Hello World: process 3 of 4, thread 6 of 8, running on comp002
Hello World: process 0 of 4, thread 2 of 8, running on comp001
```

2 nodes x
2 MPI procs/per node
= 4 MPI processes

4 MPI procs x
8 threads/proc
= 32 threads

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Question 5: How would you modify the job script to run 8 MPI processes per node and 2 OpenMP threads per MPI process?

```
#!/bin/bash
# Number of nodes requied
#SBATCH --nodes=2
# Number of tasks per node
#SBATCH --ntasks-per-node=8
# Number of CPUs per task
#SBATCH --cpus-per-task=2
# Maximum time the job will run for (HH:MM:SS)
#SBATCH --time=1:00
# Name of the job
#SBATCH --job-name=Hello_OpenMP
# Queue for the job to run in: pq is the parallel queue
#SBATCH -p pq
# Specify the account code and reservation
#SBATCH -A undergraduate_teaching-ecm3446
#SBATCH --reservation=undergraduate_teaching_2021

# Unload all exisiting modules
module purge
# Load the Intel Cluster Toolkit module used to compile the program
module load intel/2017b

# Run the program
mpirun ./hello_hybrid

# End of file
```

8 MPI processes per node

2 OpenMP threads per node

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```
Hello World: process 0 of 16, thread 0 of 2, running on comp001
Hello World: process 0 of 16, thread 1 of 2, running on comp001
Hello World: process 2 of 16, thread 0 of 2, running on comp001
Hello World: process 2 of 16, thread 1 of 2, running on comp001
Hello World: process 8 of 16, thread 0 of 2, running on comp002
Hello World: process 8 of 16, thread 1 of 2, running on comp002
Hello World: process 4 of 16, thread 0 of 2, running on comp001
Hello World: process 4 of 16, thread 1 of 2, running on comp001
Hello World: process 10 of 16, thread 0 of 2, running on comp002
Hello World: process 10 of 16, thread 1 of 2, running on comp002
Hello World: process 6 of 16, thread 0 of 2, running on comp001
Hello World: process 6 of 16, thread 1 of 2, running on comp001
Hello World: process 12 of 16, thread 0 of 2, running on comp002
Hello World: process 12 of 16, thread 1 of 2, running on comp002
Hello World: process 14 of 16, thread 0 of 2, running on comp002
Hello World: process 14 of 16, thread 1 of 2, running on comp002
Hello World: process 7 of 16, thread 0 of 2, running on comp001
Hello World: process 1 of 16, thread 0 of 2, running on comp001
Hello World: process 1 of 16, thread 1 of 2, running on comp001
Hello World: process 5 of 16, thread 0 of 2, running on comp001
Hello World: process 5 of 16, thread 1 of 2, running on comp001
Hello World: process 3 of 16, thread 0 of 2, running on comp001
Hello World: process 3 of 16, thread 1 of 2, running on comp001
Hello World: process 9 of 16, thread 0 of 2, running on comp002
Hello World: process 9 of 16, thread 1 of 2, running on comp002
Hello World: process 7 of 16, thread 1 of 2, running on comp001
Hello World: process 11 of 16, thread 0 of 2, running on comp002
Hello World: process 11 of 16, thread 1 of 2, running on comp002
Hello World: process 15 of 16, thread 0 of 2, running on comp002
Hello World: process 13 of 16, thread 0 of 2, running on comp002
Hello World: process 13 of 16, thread 1 of 2, running on comp002
Hello World: process 15 of 16, thread 1 of 2, running on comp002
```

2 nodes x
8 MPI procs/per node
= 16 MPI processes

16 MPI procs x
2 threads/proc
= 32 threads

SLURM environment variables

- Values from the SBATCH directives are propagated as environment variables. For example the hybrid job script sets:

```
SLURM_NNODES=2

SLURM_NTASKS_PER_NODE=2

SLURM_CPUS_PER_TASK=8
```

- These environment variables can be used to set up the environment if required e.g.

```
export OMP_NUM_THREADS=${SLURM_CPUS_PER_TASK}
```

Workshop 6 summary

- We have seen how to run OpenMP, MPI and hybrid programs on Isca
- The job script requests resources using directives and SLURM allocates the resources required
- Information from the SLURM directives set the number of OpenMP threads and MPI processes
- This relies on effective integration between the queuing system and the runtime environment
- For a different system consult the documentation to see how to get the best out of the system as the details can vary